BIT.MLMR

人工智能

Artificial Intelligence (AI)

2025-2026

人工智能专业　本科

know yourself






About Course

❖ Staff

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❖ Reference books

- 刘峡壁. 人工智能：机器学习与神经网络. 北京：北京理工大学出版社，2023.1
- 刘峡壁. 人工智能导论 –方法与系统. 内部教材，2026.

❖ Course website

http://www.knowyourself.xyz/AI/

- Lecture slides
- Questions
- Project solutions
- Other resources



AlphaGo

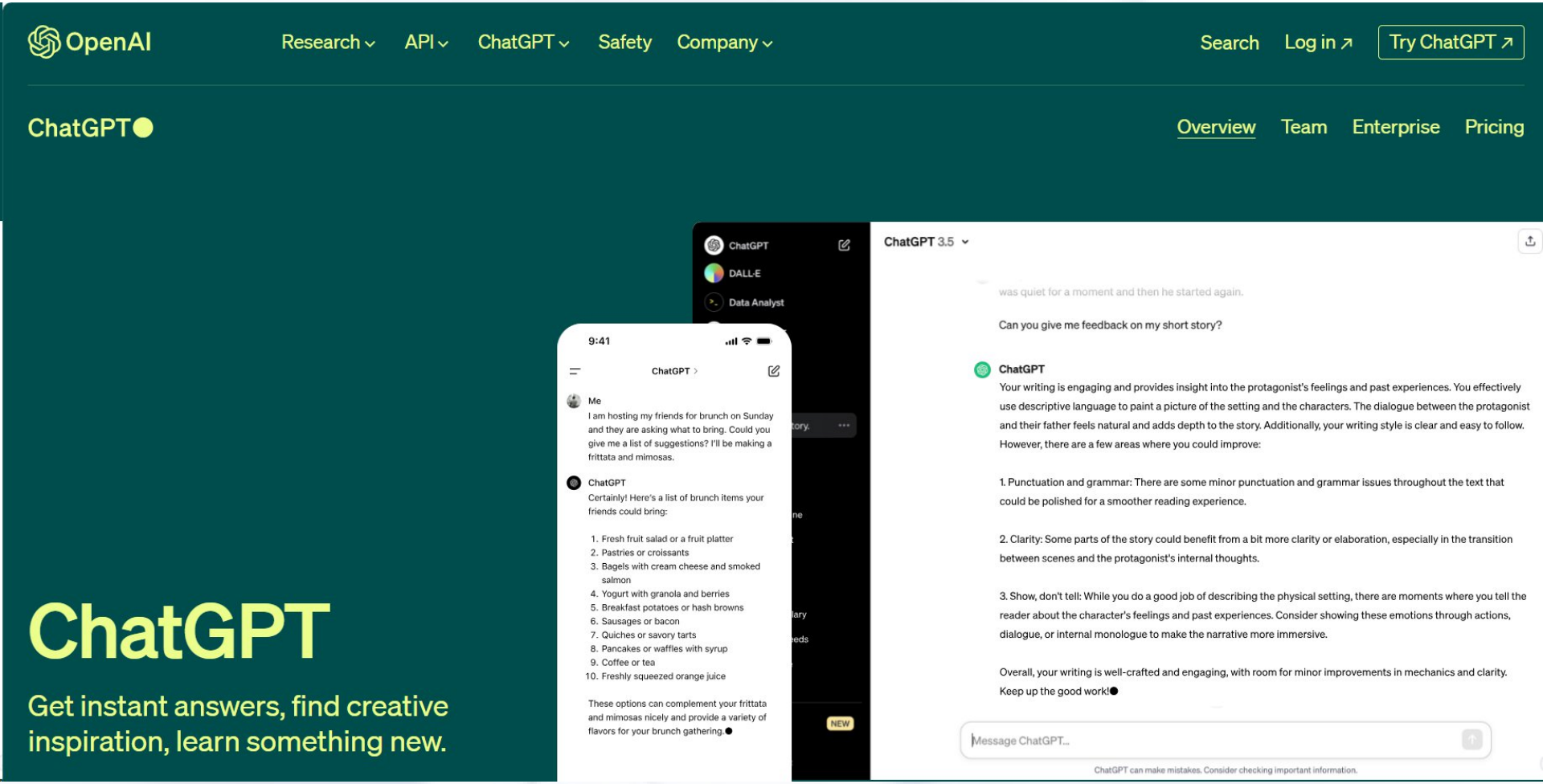
Go: AlphaGo won 5-0 in a formal match on October 2015, against the reigning 3-times European Champion, Fan Hui, becoming the first program to ever beat a professional Go player in an even game.

In March 2016 AlphaGo won 4-1 against the legendary Lee Sedol , the top Go player in the world over the past decade.





ChatGPT

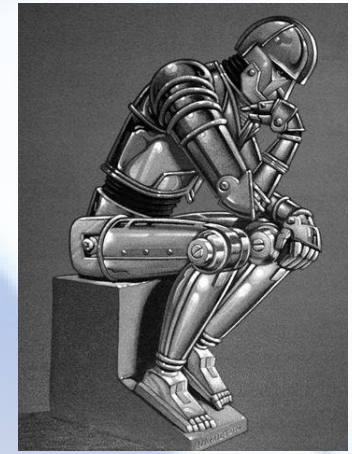

OpenAI
Research
API
ChatGPT
Safety
Company
Search
Log in ↗
Try ChatGPT ↗
ChatGPT
Overview
Team
Enterprise
Pricing
ChatGPT
Get instant answers, find creative inspiration, learn something new.
9:41
ChatGPT
Me
I am hosting my friends for brunch on Sunday and they are asking what to bring. Could you give me a list of suggestions? I'll be making a frittata and mimosas.
ChatGPT
Certainly! Here's a list of brunch items your friends could bring:
1. Fresh fruit salad or a fruit platter
2. Pastries or croissants
3. Bagels with cream cheese and smoked salmon
4. Yogurt with granola and berries
5. Breakfast potatoes or hash browns
6. Sausages or bacon
7. Quiches or savory tarts
8. Pancakes or waffles with syrup
9. Coffee or tea
10. Freshly squeezed orange juice
These options can complement your frittata and mimosas nicely and provide a variety of flavors for your brunch gathering.
ChatGPT
DALL·E
Data Analyst
NEW
ChatGPT 3.5
was quiet for a moment and then he started again.
Can you give me feedback on my short story?
ChatGPT
Your writing is engaging and provides insight into the protagonist's feelings and past experiences. You effectively use descriptive language to paint a picture of the setting and the characters. The dialogue between the protagonist and their father feels natural and adds depth to the story. Additionally, your writing style is clear and easy to follow. However, there are a few areas where you could improve:
1. Punctuation and grammar: There are some minor punctuation and grammar issues throughout the text that could be polished for a smoother reading experience.
2. Clarity: Some parts of the story could benefit from a bit more clarity or elaboration, especially in the transition between scenes and the protagonist's internal thoughts.
3. Show, don't tell: While you do a good job of describing the physical setting, there are moments where you tell the reader about the character's feelings and past experiences. Consider showing these emotions through actions, dialogue, or internal monologue to make the narrative more immersive.
Overall, your writing is well-crafted and engaging, with room for minor improvements in mechanics and clarity. Keep up the good work!
Message ChatGPT...
ChatGPT can make mistakes. Consider checking important information.

AI is always developing



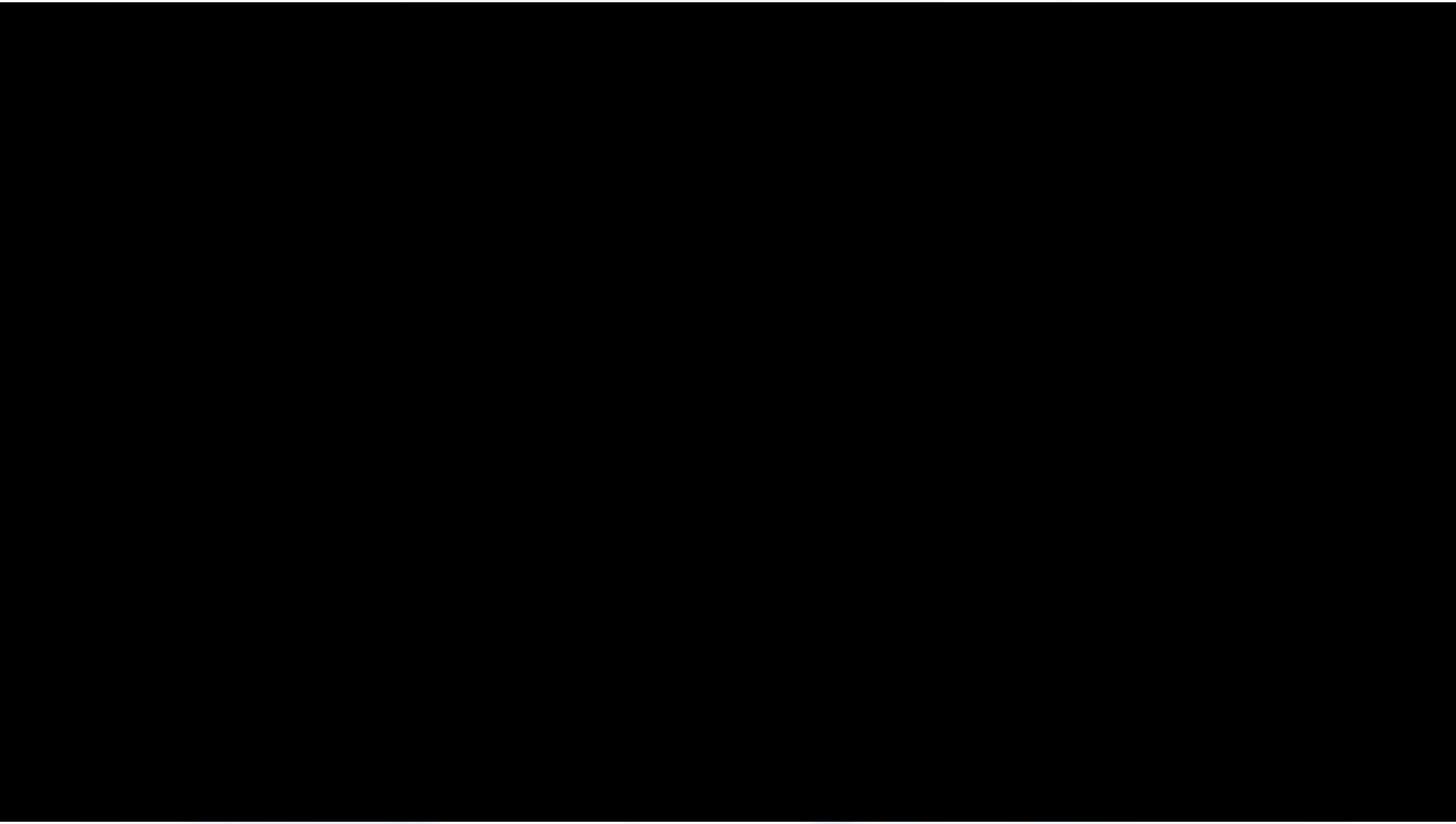
AI is always developing

Artificial Intelligence → Intelligence





What is AI reflected in movies?



What is AI reflected in movies?



What is AI reflected in movies?



Lecture Outline

- **Philosophy in Artificial Intelligence (AI)**

  What it means to think and whether artifacts could and should ever do so?

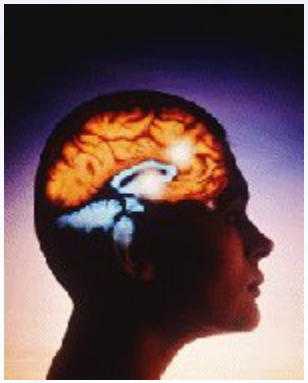
- **Ideas for AI**

  Learning, Symbolic AI, Connectionism, Nouvelle AI, Evolutionary Computation, Computational Swarm Intelligence

- **Course overview**

PartI: Philosophy in AI

What is Intelligence, anyway??



R. J. Sternberg: "*Viewed narrowly, there seem to be almost as many definitions of intelligence as there were experts asked to define it.*"

It is useful to think of intelligence in terms of an open collection of attributes.

Characteristics of Intelligence(1)



- ***Perception***

  Manipulation, integration, and interpretation of data provided by sensors, including purposeful, goal-directed, active perception



- ***Action***

  Coordination, control, and use of effectors to accomplish a variety of tasks, including exploration and manipulation of the environment, including design and construction of tools towards this end.

Characteristics of Intelligence(2)



- ***Reasoning***

  Deductive (logical) inference, inductive inference, analogical inference, hypothetical reasoning,…, including reasoning in the face of uncertainty and incomplete information.

- ***Problem-solving***

  Setting of goals (without explicit instructions from another entity), Formulation of plans, Evaluating and choosing among alternative plans, adapting plans in the face of unexpected changes

Characteristics of Intelligence(3)

- ***Learning and Adaptation***

  Learning to describe specific domains in terms of abstract theories and concepts, Learning to use, adapt, and extend language, Learning to reason, plan, and act. Adapting behavior to better cope with changing environmental demand.

- ***Sociality***

  Into social groups based on shared objectives, development of shared conventions to facilitate orderly interaction, culture.

- ***Creativity***

  Exploration, modification, and extension of domains by manipulation of domain-specific constraints, or by other means.

What is AI, anyway??

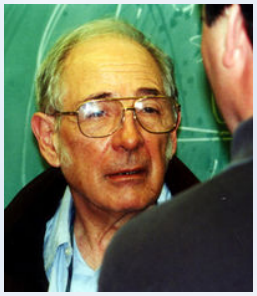
- **Understand and BUILD intelligent entities**
  - Seeking exact definition? (could last a lifetime)
- Highly interdisciplinary

  Compute Science, Philosophy, Psychology, Linguistics, NeuroScience ……
- Currently consists of huge variety of subfields

How to measure Machine Intelligence?

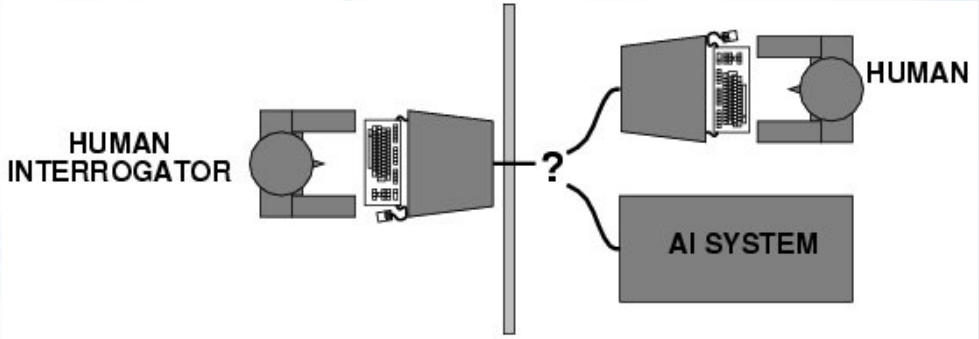
- **Two views**
  - Behavior/action (*weak AI*)
    - Can the machine act intelligently?
    - Turing test.
  - Thought process/reasoning (*strong AI*)
    - Are machines actually thinking?
    - Chinese Room of J. R. Searle





Turing test

- **When does a system behave intelligently?**
  - A. M. Turing (1950) Computing Machinery and Intelligence. Mind 49: 433-460.
  - *Operational test of intelligence*: imitation game



  - Requires the collaboration of major components of AI: knowledge, reasoning, language understanding, learning, ...

Chinese Room Argument

A man is in a room with a book of rules. Chinese sentences are passed under the door to him. The man looks up in his book of rules how to process the sentences. Eventually the rules tell him to copy some Chinese characters onto paper and pass the resulting Chinese sentences as a reply to the message he has received. The dialog continues. To follow these rules the man need not understand Chinese.

Searle, John. R. (1980) Minds, brains, and programs. *Behavioral and Brain Sciences 3 (3): 417-457*



- **Therefore, Searle says:**
  - ***the idea of a non-biological machine being intelligent is incoherent***

Chinese Room Argument



Goals of AI

- ***Current goal***

  - Making intelligent machines, especially intelligent computer programs.

  - Design and construction of useful new tools to extend human intellectual and creative capabilities

- ***Long-term goal***

  Understanding of the mechanisms underlying thought and intelligent behaviors and their embodiment in machines

Part II: Ideas for AI

Ideas for AI

- **Learning**

  ”*child machine*”

- **Connectionism**
- **Symbolic AI**
- **Evolutionary Computation**

  ”*artificial life*”

- **Computational Swarm Intelligence**
- **Nouvelle AI**

1. Learning Approach



John McCarthy:

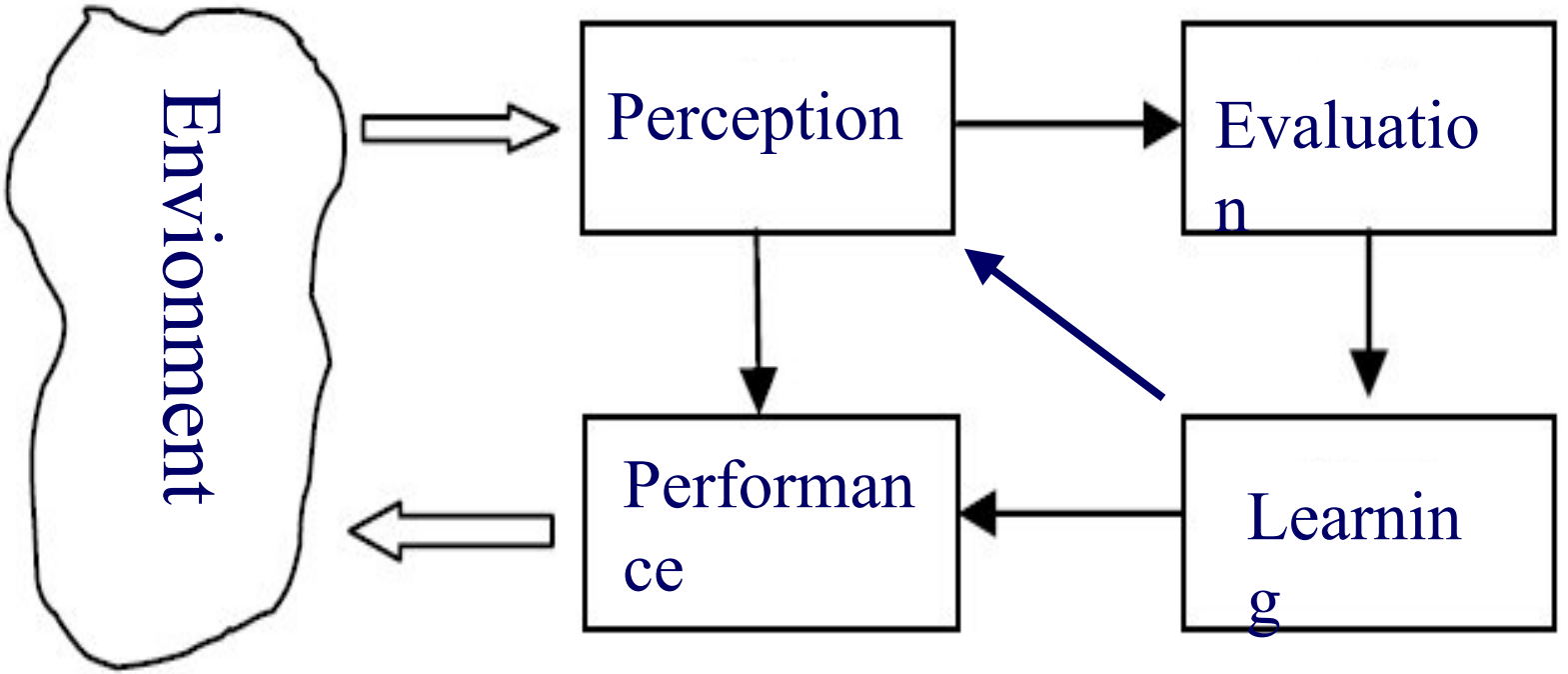
Q. What about making a ``child machine'' that could improve by reading and by learning from experience?

A. This idea has been proposed many times, starting in the 1940s. Eventually, it will be made to work. However, AI programs haven't yet reached the level of being able to learn much of what a child learns from physical experience. Nor do present programs understand language well enough to learn much by reading.

Tasks of Machine Learning

- **Learning means change**
- Improve behaviour/performance:
  - learn to perform new tasks (more)
  - increase ability on existing tasks (better)
  - increase speed on existing tasks (faster)
- Produce and increase knowledge:
  - formulate explicit concept descriptions
  - formulate explicit rules
  - discover regularities in data
  - discover the way the world behaves

The Architecture of intelligent system with learning capability



Kinds of Learning

- **Reinforcement Learning**

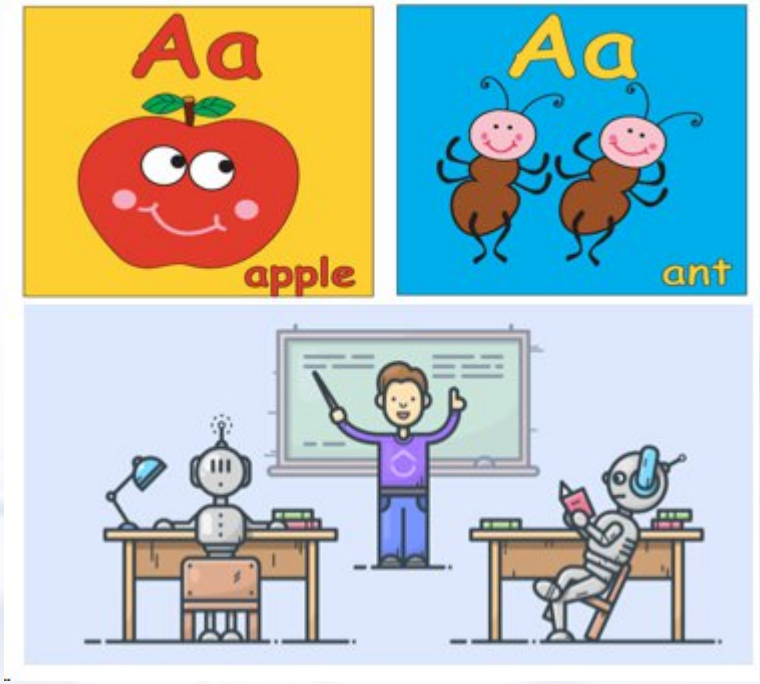
  An agent interacting with the world makes observation, takes actions, and is rewarded or punished; it should to learn to choose actions in such a way as to obtain a lot of reward.



Kinds of Learning

- **Supervised Learning**

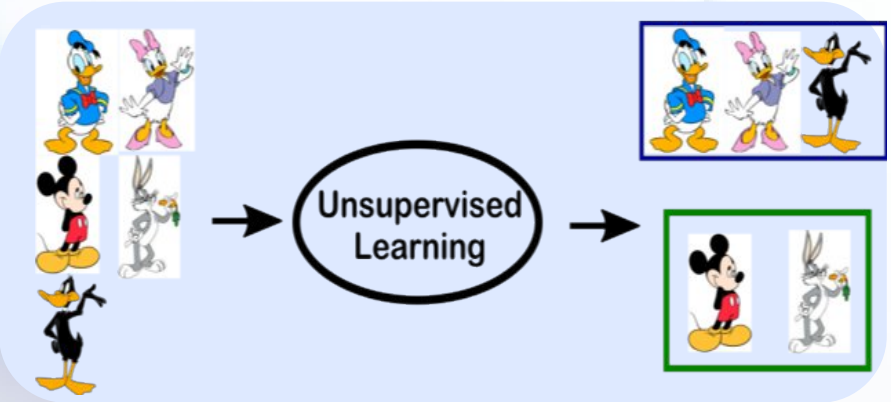
  Given a set of example input/output pairs, find a rule that does a good job or predicting the output associated with a new input.



Kinds of Learning

❖ Unsupervised Learning

Given a set of examples, no labeling of them, finding out the distribution of the data or the association rule.



clustering





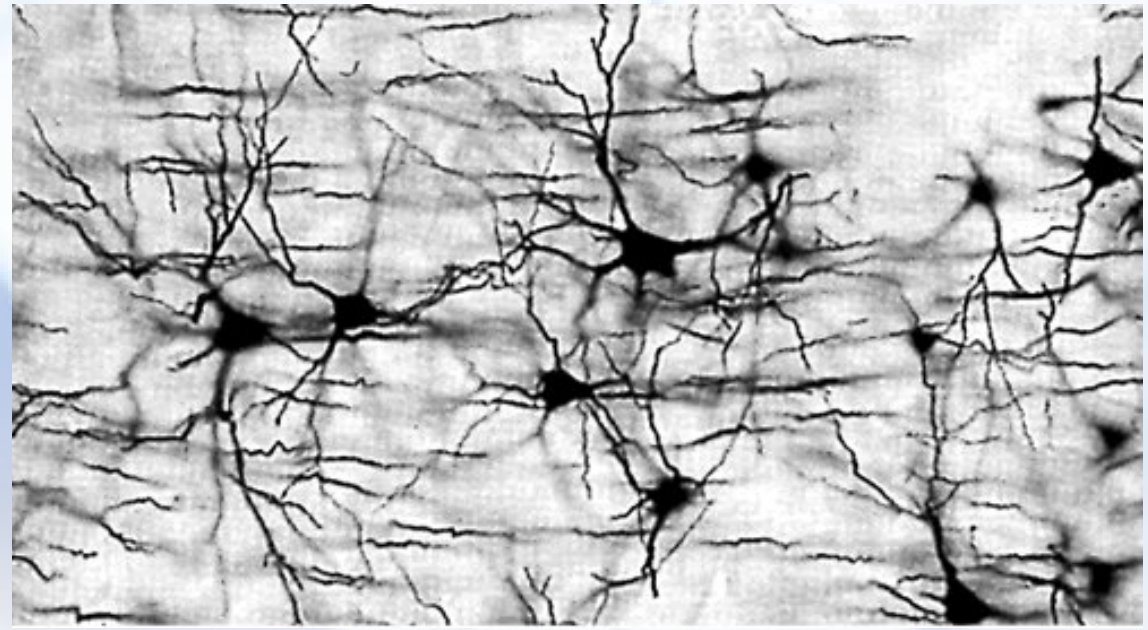
Association Rule Mining

Learning issues

- Overfitting (generalization ability) : can the machine well-trained on observed data behave well on other data either?
- Bias: which hypotheses are preferred?
- Robustness: how does the training data influence the learning result? Data Scale, Change, Noise, and Imbalance
- Computation Complexity: what is the efficiency of the learning algorithms? Time, Memory, Scalability, convergency
- Transparency: can we understand what and how has been learnt?

2. Connectionism

- **The mechanisms of brains are very different in detail from those in computers**
- **how brains work?** ☾ Bottom-up strategy



Natural Neural Network

Connectionism

- **A brief history**

  M-P neuron (McCulloch & Pitts) ☾ Perceptron (Rosenblatt) ☾ Hopfield Model, B-P Learning Method (Rumelhart & McClelland) ☾ Deep Learning (Geoffrey Hinton)

- **Applications**

  Recognition, Vision, Business, Medical, …….

- **Core Issues**

  - Topology

  - Learning Methods

AlphaGo - Ground-breaking



Artificial Brain

- Artificial brains are a man-made machines that have the same cognitive ability as humans and other mammals.
- Projects

  SyNAPSE: DAPRA, with IBM, HP, HRL Labs.

  Blue Brain: EPFL Together with IBM

  Barin in Silicon: Standford University

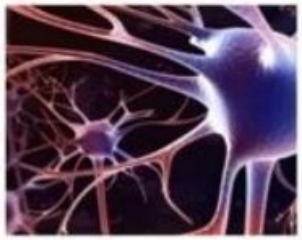
  .....

Ref: http://www.artificialbrains.com/

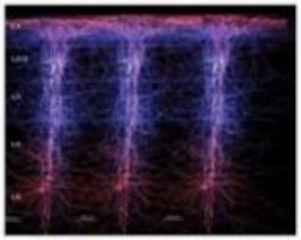
Neuromorphic chips for the Artificial Brain



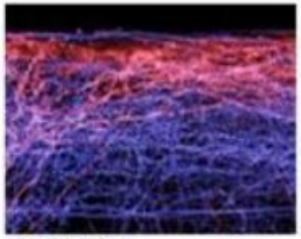
~10^{10} synapses/cm^2



~10^6 neurons/cm^2



~10^4 neurons/ cortical column



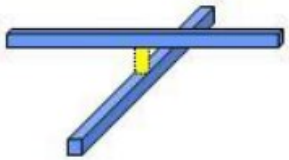
~5X10^8 long range axons @ 1 Hz



Biological Brain

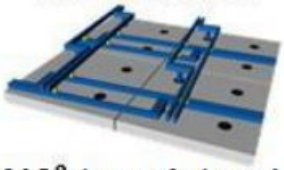
Synapse | Neurons | Microcircuit | Long-range interconnects | Brain

CROSSBAR JUNCTION



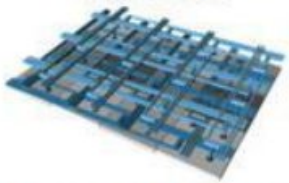
~10^{10} intersections/ cm^2 @ 100 nm pitch

CMOS SUBSTRATE



5X10^8 transistors/ cm^2 @ 500 transistors/ neuron

LAMINAR CIRCUIT

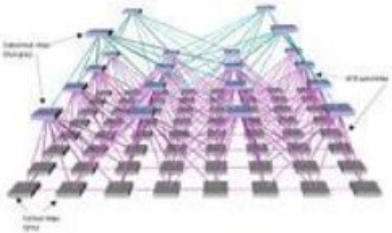


Layered cortical circuits with ~10^6 neurons/cm^2

HIGH SPEED BUS



Multi-Gbit/sec digital comms



Electronic Brain

3. Symbolic AI

- ***Physical Symbol System Hypothesis*** of Newell and Simon

  - the processing of structures of symbols by a digital computer is sufficient to produce artificial intelligence

  - the processing of structures of symbols by the human brain is the basis of human intelligence

  - it remains an open question whether the Physical Symbol System Hypothesis is true or false

  - Top-down strategy

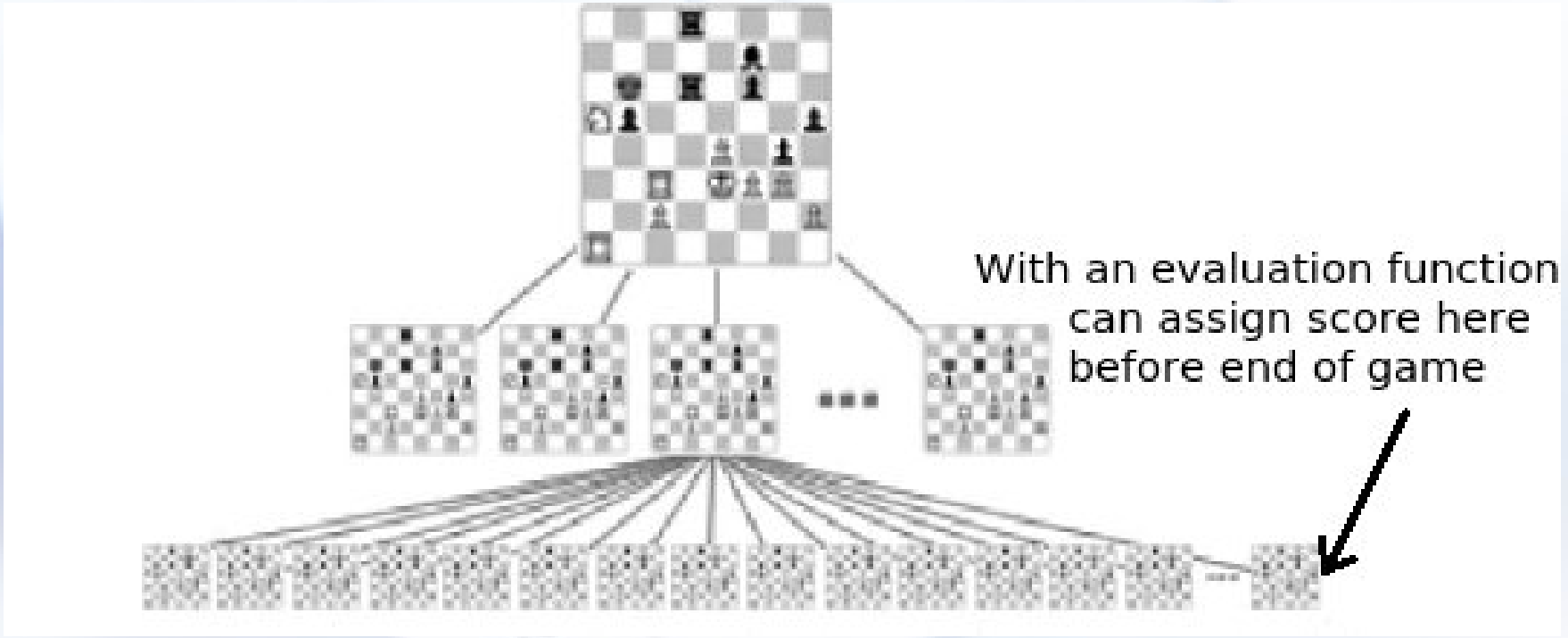
Symbolic AI

- **Problem-sloving ☾ Expert System ☾ Knowledge Engineering**
  - **Search, Representation, Reasoning**
  - **GPS, Deep Blue, DENDRAL, CYC…..**



Search Problem

How to search is a key to Symbolic AI as well as AI



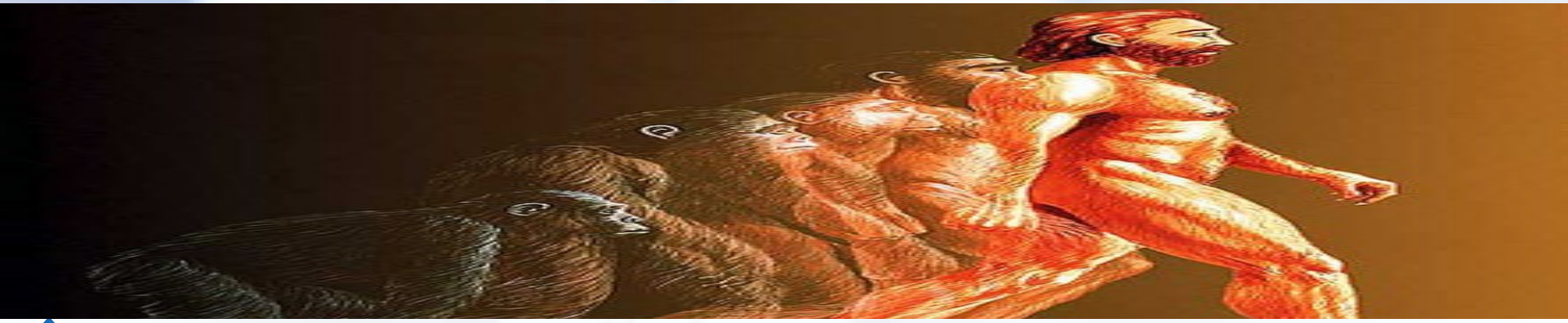
4. Evolutionary Computation

- ***Biological evolution***

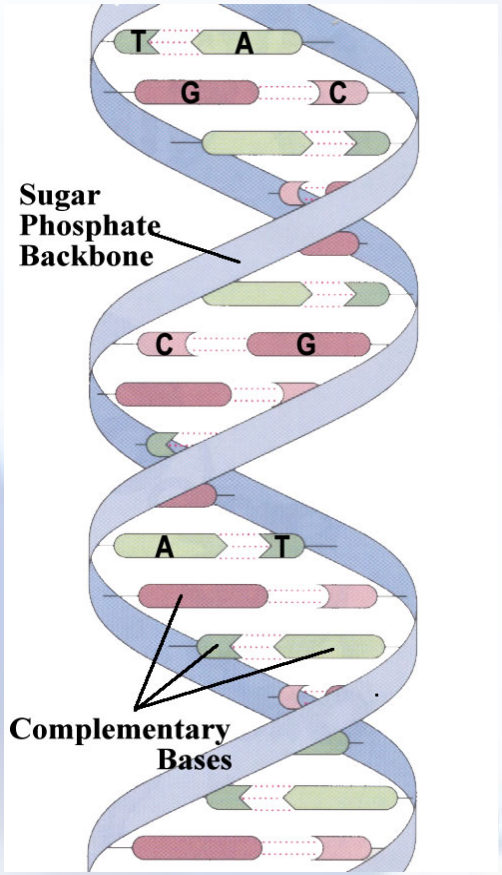
  To produce an enormous variety of living organisms closely suited to different sets of needs in different environments.

- ***Simulated evolution***

  By modeling those processes of biological evolution on computers, it turns out that we can sometimes get the computers to evolve solutions to problems.



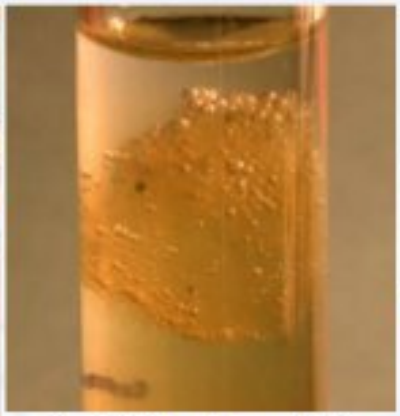
DNA Computing



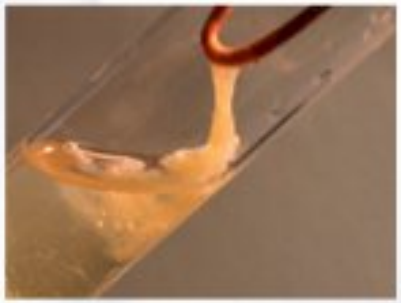
A T G G A T C A G C T G A
T A C C T A G T C G A C T



Formation of DNA strands.



Precipitation of more DNA strands in alcohol



Spooling the DNA with a metal hook or similar device

Evolutionary Computation

- Genetic Algorithm

  Use *strings of symbols* to encode solutions to problems, like strings of molecules in DNA. *Transforming* and *recombining* portions of strings enables an evolutionary computation to search for good solutions, partly analogous to biological evolution.

  Genetic Programming

  Extends these ideas to automatic programming by using *structures* which are better suited to the problem than strings are.

Evolutionary Computation

- ## Evolutionary Strategy

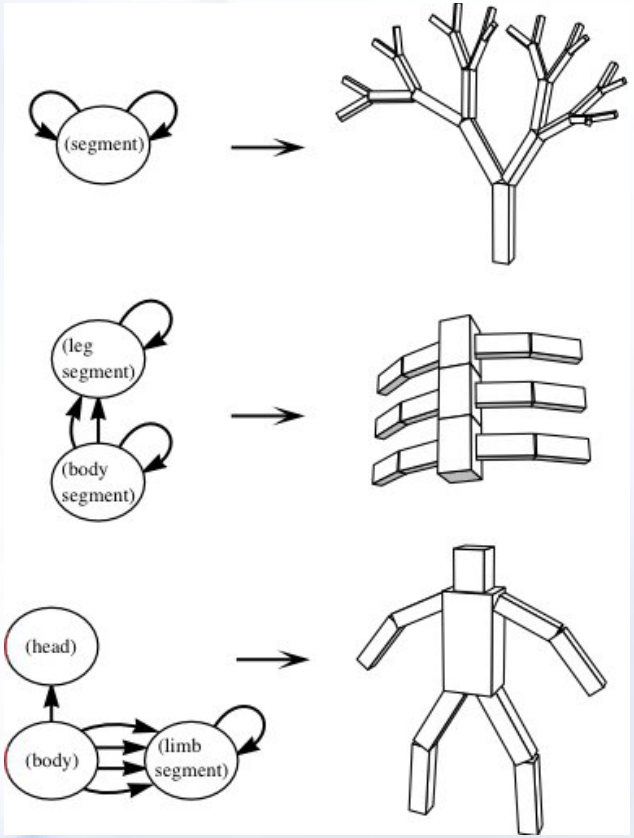
  Use *natural problem-dependent representations*, and primarily mutation and selection as search operators. *Mutation* is normally performed by adding a normally distributed random value to each vector component. The step size or mutation strength is often governed by self-adaptation. The *selection* in evolution strategies is deterministic and only based on the fitness rankings, not on the actual fitness values.

- ## Evolutionary Programming

  Harder to distinguish from evolutionary strategies. Its main variation operator is *mutation*; members of the population are viewed as part of *a specific species* rather than members of the same species therefore each parent generates an offspring.

Example: Forming body plans with evolution

- Node specifies part type, joint, and range of movement
- Edges specify the joints between parts
- Population?
  - Graphs of nodes and edges
- Selection?
  - Ability to perform some task (walking, jumping, etc.)
- Mutation?
  - Node types change/new nodes grafted on



From Virgil Griffith, Google Tech Talk - 2007

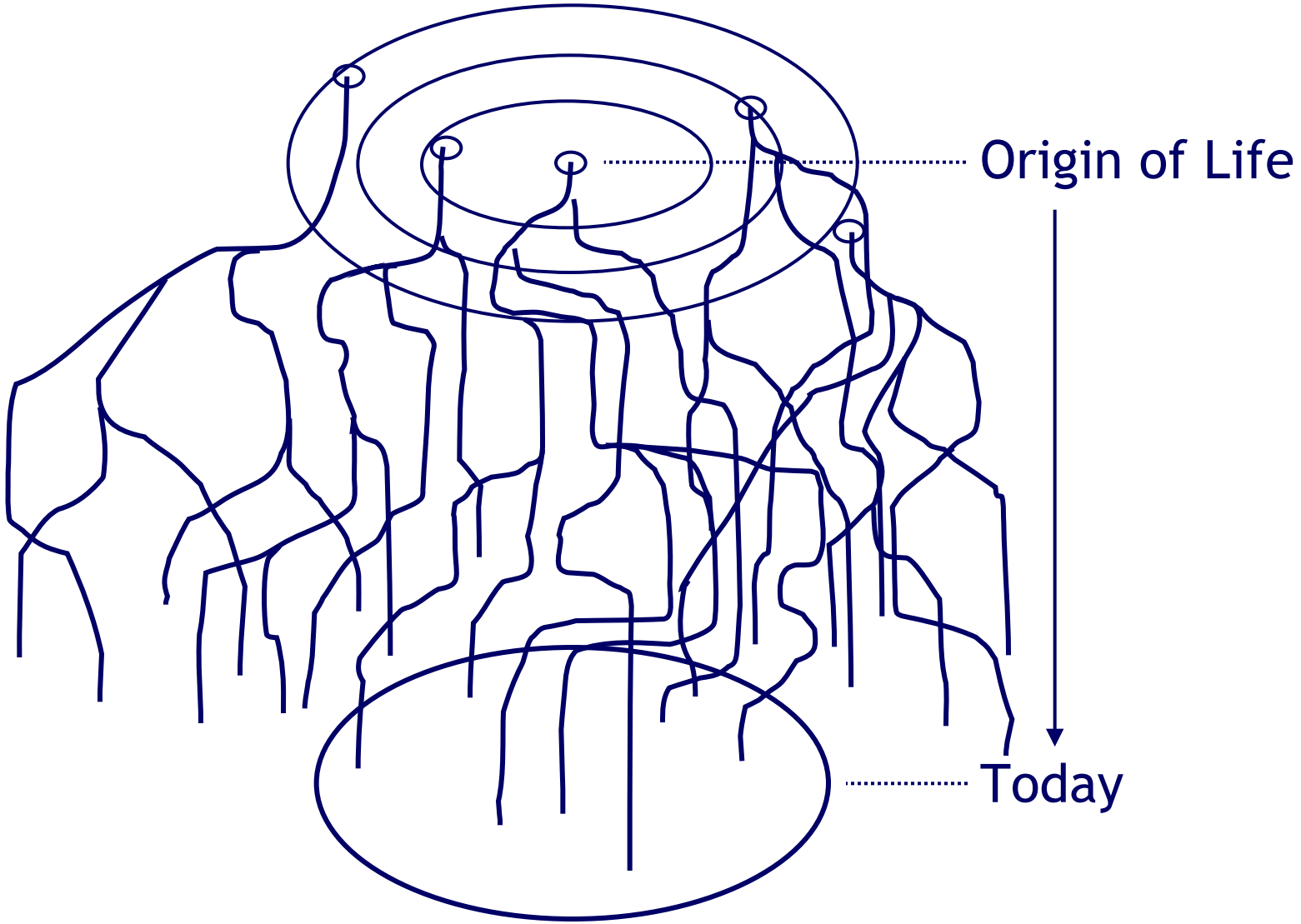
Artificial Life (Alife)

- Artificial Life is the study of man-made systems that exhibit behaviors characteristic of natural living systems. It complements the traditional biological sciences concerned with the *analysis* of living organisms by attempting to *synthesize* life-like behaviors within computers and other artificial media. By extending the empirical foundation upon which biology is based *beyond* the carbon-chain life that has evolved on Earth, Artificial Life can contribute to theoretical biology by locating *life-as-we-know-it* within the larger picture of *life-as-it-could-be."*

  Chris Langton (in Proc. of first Alife conference)

Artificial Life and Evolutionary

Life, as it is… and might have been



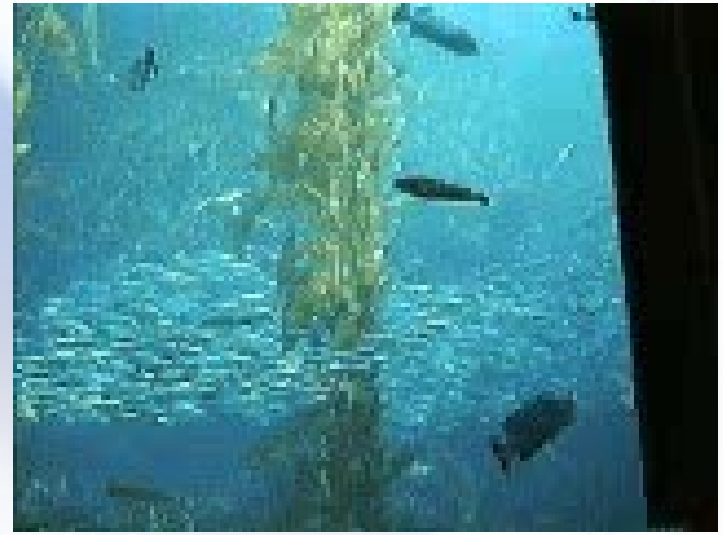
5. Computational Swarm Intelligence

- Intelligence is often considered a property of *individuals*.
- Are we social because we are intelligent or are we intelligent because we are social?
  - Intelligence can emerge from social interaction.
- *Emergent behaviour* – when a group behaves in ways that were not ”programmed” into its members.
- *Swarm intelligence*
  - simulated social interaction
  - emergent collective intelligence of groups of simple agents

Computational Beauty in Nature









Observations

- Bird flocks and fish schools move in a coordinated way, but there is no coordinator (leader)

  - So, what decides the behaviour of a leader-less flock?

- Ants and termites quickly find the shortest path between the nest and a food source

  - ... and solve many other advanced problems as well: keeping cattle, building (ventilated) housing, coordinated heavy transports, tactical warfare, cleaning house, etc.

  - A single ant is essentially a blind, memory-less, random walker!

- Distributed systems without central control
- Useful not only to simulate but also to solve optimization problems

Computational Simulation

- *Multi-Agent Systems*
  - a system composed of multiple interacting intelligent agents.
  - application including computer games, networks, transportation, logistics, and etc.
- *Ant Colony Optimization*
  - 1991 (Dorigo)
  - mostly for combinatorial optimization
- *Particle Swarm Optimization*
  - 1995 (Kennedy & Eberhart)
  - more general optimization technique

6. Nouvelle AI

- **Rodney Brooks (1991)**

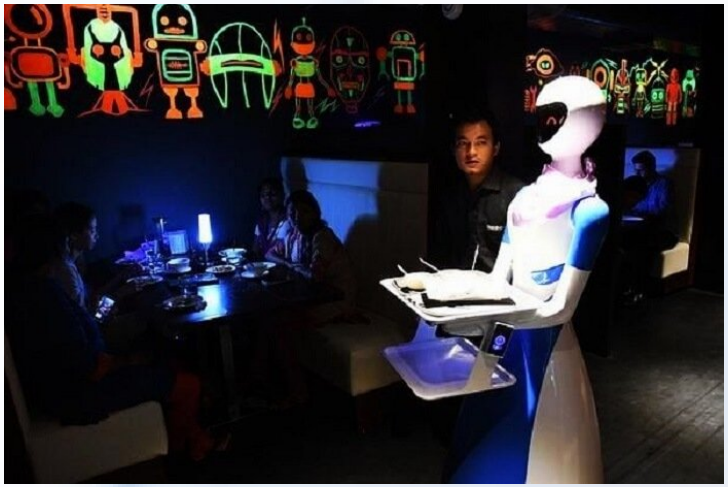
  <<Intelligence without Representation>>

  Insect-like mobile robots: Allen, Herbert, Genghis

- The basic building blocks of intelligence are very simple behaviours, More complex behaviours "emerge" from the interaction of these simple behaviours.
- Producing systems that display approximately the same level of intelligence as insects.

Nouvelle AI

- **Situated AI**
  - Build disembodied intelligences who unfriendly interact with the world (traditional)
  - Build embodied intelligences situated in a real world (Nouvelle).



- **Lifetime Learning**
  - Reinforcement learning
  - Adapt to environment by acting and receiving reward/punishment in the environment.

Course Overview

PartI Machine Learning

Concepts, Methods, Supervised and Unsupervised Learning

Part II Connectionism

Concepts, Problems, Models

PartII Symbolic AI

Problem representation, Graph Search, Adversarial Search, Reasoning

PartV Evolutionary Computation

Genetic Algorithms, Evolutionary Programming, Evolutionary Strategies

PartVI Computational Swarm Intelligence

Ant Colony Optimization, Particle Swarm Optimization

PartIV Nouvelle AI

Agent, Reinforcement Learning

GOAL!!!

Get a general understanding of AI,

preparing yourself for learning and study of braches of AI!

Grading

Final exam: 40%

Term projects: 50%

Participation in the class and Exercises: 10%

Bonus: at most 12 scores

Learning to learn

Loris Malaguzzi: Learning and teaching should not stand on opposite banks and just watch the river flow by; instead, they should embark together on a journey down the water. Through an active, reciprocal exchange, teaching can strengthen learning how to learn.

BIT.MLMR

End of Lecture1

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